

Course 2001A

Semester II

Theory · 4 Credits

3:1:0:0

# Engineering Mathematics – I

Complex Numbers · Probability & Statistics · Integral Calculus · Numerical Methods & Vector Algebra

Complete handbook for the Kerala Diploma Revision 2026 curriculum. Covers all four modules with deep explanations, worked examples, interactive calculators, Malayalam support, and exam-oriented practice.

|               |                                              |
|---------------|----------------------------------------------|
| PROGRAMME     | Common to all Engineering Diploma programmes |
| COURSE TYPE   | Theory                                       |
| CONTACT HOURS | 60 (L:45 + T:15)                             |
| SELF-LEARNING | 60 hours (suggested)                         |
| CIE / ESE     | 40 / 60                                      |

|            |                                           |
|------------|-------------------------------------------|
| MODULE I   | Complex Numbers (14 h)                    |
| MODULE II  | Probability & Statistics (15 h)           |
| MODULE III | Integral Calculus (15 h)                  |
| MODULE IV  | Numerical Methods & Vector Algebra (16 h) |



# Official Source Declaration

**Source:** Kerala Polytechnic Revision 2026 Syllabus — Course 2001A (Engineering Mathematics – I).

This handbook is an educational study aid developed for students. It is not an official publication of the examining body.

**⚠ Verified Inconsistencies:** The syllabus lists Module IV self-learning activities with a typo — "simpson's  $\left(\frac{1}{2}\right)$  rd rule" — which is corrected to **Simpson's 1/3rd rule** throughout this handbook, matching the main syllabus entry. The total self-learning hours sum to 60 as stated. All other data reconcile.



## Course Dashboard

60

CONTACT HOURS

4

CREDITS

40+60

CIE / ESE

4

MODULES

### Prerequisites

**Fundamentals of Engineering Mathematics I** (Course 1002) — Trigonometry, Determinants, Differentiation.

### Teaching & Assessment Scheme

| Teaching Scheme   | Self-Learning/Week | Total Hours | Credits | CIE | ESE |
|-------------------|--------------------|-------------|---------|-----|-----|
| 3:1:0:0 (L:T:P:R) | 4 h                | 60          | 4       | 40  | 60  |

| Module-wise Instructional Hours |                                    |             |              |       |
|---------------------------------|------------------------------------|-------------|--------------|-------|
| Module                          | Title                              | Lecture (L) | Tutorial (T) | Total |
| I                               | Complex Numbers                    | 10          | 4            | 14    |
| II                              | Probability & Statistics           | 12          | 3            | 15    |
| III                             | Integral Calculus                  | 11          | 4            | 15    |
| IV                              | Numerical Methods & Vector Algebra | 12          | 4            | 16    |
| Total                           |                                    |             | 60           |       |

## How to Use This Handbook

1

### Understand

Read each module's explanation, formulae, and worked examples.

2

### Visualise

Study the SVG diagrams and interactive calculators.

3

### Apply

Solve the module-wise questions and try the model paper.

4

### Revise

Use the Quick Revision cards and glossary before exams.

## Course Outcomes

## CO1

Apply concepts of complex numbers to solve problems. (Bloom: Apply)

## CO2

Apply concepts of Probability & Statistics in various mathematical problems. (Bloom: Apply)

## CO3

Use integration techniques and differential equations to solve basic engineering problems. (Bloom: Apply)

## CO4

Apply basic numerical methods and vector algebra techniques to solve simple engineering problems. (Bloom: Apply)

### CO–PO Mapping

| CO  | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | -   | -   | -   | -   | -   | -   | -   | -   | -    | -    |
| CO2 | 3   | -   | -   | -   | -   | -   | -   | -   | -   | -    | -    |
| CO3 | 3   | -   | -   | -   | -   | -   | -   | -   | -   | -    | -    |
| CO4 | 3   | -   | -   | -   | -   | -   | -   | -   | -   | -    | -    |

Legends: 1–Low; 2–Medium; 3–High; – No Correlation



## Curriculum Coverage Map

### Module → Topics → Hours → CO → Handbook Anchor

| Module | Major Topics                                                                                                | Hours | CO  | Section  |
|--------|-------------------------------------------------------------------------------------------------------------|-------|-----|----------|
| I      | Complex number system, operations, Argand plane, modulus, amplitude, polar form, problems                   | 14    | CO1 | Module 1 |
| II     | Sets, probability axioms, coins/dice/cards, statistics, frequency distribution, mean/median/mode            | 15    | CO2 | Module 2 |
| III    | Integration as reverse, standard integrals, substitution, parts, definite integrals, differential equations | 15    | CO3 | Module 3 |
| IV     | Interpolation, Lagrange, Simpson's 1/3rd, scalars/vectors, dot/cross product, magnitude, unit vector        | 16    | CO4 | Module 4 |



## How the Modules Connect

## Module 1

### Complex Numbers

Foundation for AC circuit analysis, signal processing, and control systems.

## Module 2

### Probability & Stats

Data analysis, quality control, risk assessment, and decision-making.

## Module 3

### Integral Calculus

Area, volume, work, and modelling dynamic systems via differential equations.

## Module 4

### Numerical Methods & Vectors

Approximation, numerical integration, and spatial/physical vector analysis.

## MODULE I **Complex Numbers**

14 hours (L:10 + T:4)

CO1

Bloom: Apply

## Learning Goals

- Understand the complex number system
- Perform fundamental operations
- Represent complex numbers graphically
- Find modulus, amplitude, and polar form
- Solve engineering problems

## Key Facts

- $i = \sqrt{-1}$
- $z = a + ib$
- $|z| = \sqrt{a^2 + b^2}$
- $\arg(z) = \tan^{-1}(b/a)$
- Polar:  $z = r(\cos \theta + i \sin \theta)$

## The Complex Number System

A **complex number** is a number of the form  $z = a + ib$ , where **a** and **b** are real numbers, and **i** is the imaginary unit defined by  $i^2 = -1$ .

- **Real part:**  $\text{Re}(z) = a$
- **Imaginary part:**  $\text{Im}(z) = b$

$$z = a + ib \cdot a, b \in \mathbb{R} \cdot i^2 = -1$$

**Example:**  $z = 3 + 4i$  has  $\text{Re} = 3$  and  $\text{Im} = 4$ .

**Malayalam support:** കോംപ്ലക്സ് സംഖ്യകൾ (complex numbers)  $a + ib$  രൂപത്തിൽ എഴുതാം. ഇവിടെ 'i'  $\sqrt{-1}$  നെ സൂചിപ്പിക്കുന്നു.  $i^2 = -1$  ആണ്.  $a$  യും  $b$  യും യഥാക്രമേ  $\text{Re}(z)$  (Real part) ഉം  $\text{Im}(z)$  (Imaginary part) ഉമാണ്.

 **Tip:** The real and imaginary parts are always real numbers. The imaginary part includes the coefficient of  $i$ , not  $i$  itself.

## Conjugate and Equality

**Conjugate:** The conjugate of  $z = a + ib$  is  $\bar{z} = a - ib$ .

$$\bar{z} = a - ib$$

**Equality:** Two complex numbers  $z_1 = a_1 + ib_1$  and  $z_2 = a_2 + ib_2$  are equal iff  $a_1 = a_2$  **and**  $b_1 = b_2$ .

**Example:** If  $2x + 3iy = 4 + 6i$ , then  $2x = 4 \Rightarrow x = 2$ , and  $3y = 6 \Rightarrow y = 2$ .



**Tip:** The conjugate is used to rationalise denominators and find modulus.

## Fundamental Operations

For  $z_1 = a + ib$  and  $z_2 = c + id$ :

- **Addition:**  $z_1 + z_2 = (a+c) + i(b+d)$
- **Subtraction:**  $z_1 - z_2 = (a-c) + i(b-d)$
- **Multiplication:**  $z_1 \cdot z_2 = (ac - bd) + i(ad + bc)$
- **Division:**  $z_1 / z_2 = (a+ib)/(c+id) \times (c-id)/(c-id) = [(ac+bd) + i(bc-ad)] / (c^2 + d^2)$

**Example (Multiplication):**  $(2+3i)(1-4i) = 2 - 8i + 3i - 12i^2 = 2 - 5i + 12 = 14 - 5i$

**Example (Division):**  $(3+4i)/(1-2i) = (3+4i)(1+2i)/5 = (3+6i+4i+8i^2)/5 = (-5+10i)/5 = -1 + 2i$



**Common mistake:** In division, always multiply numerator and denominator by the conjugate of the denominator.

## Graphical Representation — Argand Plane

Complex numbers are represented on the **Argand plane** (complex plane) with the real part on the horizontal axis and the imaginary part on the vertical axis.

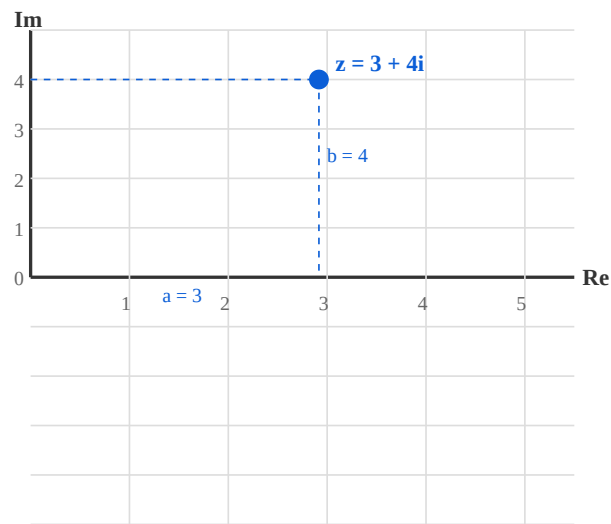


Figure 1: Argand plane showing  $z = 3 + 4i$  with real part  $a = 3$  and imaginary part  $b = 4$ .

**Malayalam support:** Argand plane- complex numbers- points represent . Horizontal axis real part-um vertical axis imaginary part-um .

## Modulus and Amplitude

**Modulus (Absolute value):**  $|z| = \sqrt{a^2 + b^2}$ . It is the distance from the origin to the point  $(a, b)$ .

**Amplitude (Argument):**  $\theta = \arg(z) = \tan^{-1}(b/a)$ , measured in degrees (or radians).

$$|z| = \sqrt{a^2 + b^2} \cdot \theta = \tan^{-1}(b/a)$$

**Example:** For  $z = 3 + 4i$ :  $|z| = \sqrt{9+16} = 5$ ,  $\theta = \tan^{-1}(4/3) \approx 53.13^\circ$

**! Common mistake:** The argument depends on the quadrant. Always check the signs of  $a$  and  $b$ . In QII,  $\theta = 180^\circ - \tan^{-1}(b/|a|)$ .



## Polar Form

A complex number can be expressed in **polar form** as:

$$z = r(\cos \theta + i \sin \theta) \cdot r = |z|, \theta = \arg(z)$$

This is also written as  $z = r \angle \theta$  or  $z = r \operatorname{cis} \theta$ .

**Example:** For  $z = 3 + 4i$ :  $r = 5$ ,  $\theta = 53.13^\circ \Rightarrow z = 5(\cos 53.13^\circ + i \sin 53.13^\circ)$

 **Tip:** Polar form is especially useful for multiplication and division — multiply moduli and add arguments.

## Solved Problems

**Problem 1:** Find the modulus and argument of  $z = -1 + i$ .

Solution:  $a = -1$ ,  $b = 1$ .  $r = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$ .  $\theta = 180^\circ - \tan^{-1}(1/1) = 180^\circ - 45^\circ = 135^\circ$ .

**Problem 2:** Express  $z = 2 - 2i$  in polar form.

Solution:  $r = \sqrt{2^2 + (-2)^2} = 2\sqrt{2}$ .  $\theta = -\tan^{-1}(2/2) = -45^\circ$ .  $z = 2\sqrt{2}(\cos(-45^\circ) + i \sin(-45^\circ))$ .

**Problem 3:** If  $z_1 = 1+2i$  and  $z_2 = 3-i$ , find  $z_1 \times z_2$ .

Solution:  $(1+2i)(3-i) = 3 - i + 6i - 2i^2 = 3 + 5i + 2 = 5 + 5i$ .

 **Engineering Application:** Complex numbers model AC circuits — impedance  $Z = R + iX$ , with magnitude giving total opposition and phase angle giving phase shift.



## Complex Number Calculator

Enter real and imaginary parts to compute modulus, argument, and polar form.

Real part (a)

Imaginary part (b)

3

4

$z = 3 + 4i$  |  $|z| = 5.000$  |  $\arg = 53.13^\circ$  | Polar:  $5.000(\cos 53.13^\circ + i \sin 53.13^\circ)$

## MODULE II Probability & Statistics

15 hours (L:12 + T:3)

CO2

Bloom: Apply



### Learning Goals

- Understand sets, random experiments, sample space
- Apply classical probability axioms
- Solve problems with coins, dice, cards
- Organise data and compute mean/median/mode



### Key Facts

- $P(E) = n(E)/n(S)$
- $0 \leq P(E) \leq 1$
- Mean =  $\Sigma x/n$
- Median = middle value (ordered)
- Mode = most frequent value

## Sets: Empty, Universal, Subsets

- **Set:** A well-defined collection of objects.
- **Empty set ( $\emptyset$ ):** A set with no elements.
- **Universal set (U):** The set containing all elements under consideration.
- **Subset:**  $A \subseteq B$  if every element of A is also in B.

**Example:** For rolling a die,  $U = \{1, 2, 3, 4, 5, 6\}$ ,  $A = \{2, 4, 6\}$  (even numbers) is a subset of U.

**Malayalam support:** Set എന്നത് നിശ്ചിതമായ വസ്തുക്കളുടെ ശേഖരമാണ്. Empty set-ന് എന്തും ഉണ്ടാകില്ല. Universal set-ന് എന്തും ഉണ്ടാകാം.

## Random Experiments, Outcomes, Sample Space

- **Random experiment:** An experiment whose outcome is not certain.
- **Outcome:** A single result of an experiment.
- **Sample space (S):** The set of all possible outcomes.
- **Event (E):** A subset of the sample space.

**Example:** Tossing a coin:  $S = \{H, T\}$ . Event  $E = \{H\}$  (getting heads).

## Classical Definition & Axioms of Probability

**Classical definition:** For a random experiment with  $n(S)$  equally likely outcomes, the probability of event E is:

$$P(E) = n(E) / n(S)$$

**Axioms:**

- $0 \leq P(E) \leq 1$
- $P(S) = 1$
- For mutually exclusive events,  $P(E_1 \cup E_2) = P(E_1) + P(E_2)$



**Tip:** The sum of probabilities of all outcomes in S is 1.

## Problems: Coins, Die, Cards

**Coin (1 coin):**  $S = \{H, T\}$ .  $P(H) = 1/2$ .

**Coin (2 coins):**  $S = \{HH, HT, TH, TT\}$ .  $P(HT) = 1/4$ ,  $P(\text{at least one H}) = 3/4$ .

**Die:**  $S = \{1, 2, 3, 4, 5, 6\}$ .  $P(\text{even}) = 3/6 = 1/2$ .

**Cards:** From a deck of 52,  $P(\text{heart}) = 13/52 = 1/4$ .

 **Common mistake:** In "at least one" problems, use the complement rule:  $P(\text{at least one}) = 1 - P(\text{none})$ .

## Introduction to Statistics

**Statistics:** The science of collecting, organising, analysing, and interpreting data.

- **Ungrouped data:** Raw data as collected.
- **Frequency distribution:** A table showing how often each value occurs.

**Example:** Marks: 5, 7, 7, 8, 9. Frequency: 5→1, 7→2, 8→1, 9→1.

## Measures of Central Tendency

**Mean ( $\bar{x}$ ):** Sum of all values  $\div$  number of values.

$$\bar{x} = \Sigma x / n$$

**Median:** The middle value when data is ordered (or average of two middle values if  $n$  is even).

**Mode:** The value that occurs most frequently.

**Example:** Data: 3, 5, 7, 7, 9. Mean =  $31/5 = 6.2$ . Median = 7. Mode = 7.

**Example (even  $n$ ):** 2, 4, 6, 8. Median =  $(4+6)/2 = 5$ .



**Tip:** Mean is affected by outliers; median and mode are more robust.

## Frequency Distribution — Mean, Median, Mode

For grouped frequency distribution:

$$\bar{x} = \Sigma(f \cdot x) / \Sigma f$$

**Median** for grouped data: find the class containing  $(n+1)/2$ -th observation.

**Mode** for grouped data: the class with highest frequency.

**Example:**  $x$ : 1, 2, 3, 4     $f$ : 2, 3, 1, 2.  $\Sigma f = 8$ .  $\Sigma(f \cdot x) = 2+6+3+8 = 19$ .  $\bar{x} = 19/8 = 2.375$ .

**Malayalam support:** Frequency distribution- $\bar{x}$  data- $\bar{x}$  groups  $\bar{x}$   $\bar{x}$  frequencies  $\bar{x}$   $\bar{x}$ . Mean, median, mode  $\bar{x}$  central tendency- $\bar{x}$  measures  $\bar{x}$ .



### Mean, Median, Mode Calculator

Enter numbers separated by commas (e.g. 5,7,7,8,9).

Data

5,7,7,8,9

Mean: 7.200 | Median: 7.000 | Mode: 7

## MODULE III Integral Calculus

15 hours (L:11 + T:4)

CO3

Bloom: Apply



### Learning Goals

- Understand integration as reverse of differentiation
- Recall standard integrals
- Apply substitution and integration by parts
- Evaluate definite integrals
- Solve first-order differential equations



### Key Facts

- $\int x^n dx = x^{n+1}/(n+1) + C$  ( $n \neq -1$ )
- $\int e^x dx = e^x + C$
- $\int 1/x dx = \ln|x| + C$
- $\int \sin x dx = -\cos x + C$
- $\int \cos x dx = \sin x + C$

## Integration as Reverse of Differentiation

**Integration** is the inverse process of differentiation. If  $\frac{d}{dx} [F(x)] = f(x)$ , then  $\int f(x) dx = F(x) + C$ , where  $C$  is the constant of integration.

$$\int f(x) dx = F(x) + C \quad \cdot \quad \frac{d}{dx} [F(x)] = f(x)$$

**Example:**  $\frac{d}{dx} (x^2) = 2x \Rightarrow \int 2x dx = x^2 + C.$



**Tip:** The constant  $C$  accounts for all antiderivatives. Always include it in indefinite integrals.

## Standard Integrals

$$\int x^n dx = x^{n+1}/(n+1) + C$$

$$n \neq -1$$

$$\int 1/x dx = \ln|x| + C$$

$$x \neq 0$$

$$\int e^x dx = e^x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$$

## Rules of Integration & Substitution

### Rules:

- $\int k \cdot f(x) \, dx = k \cdot \int f(x) \, dx$
- $\int [f(x) \pm g(x)] \, dx = \int f(x) \, dx \pm \int g(x) \, dx$

### Substitution:

- $\int f(ax+b) \, dx = (1/a) F(ax+b) + C$
- $\int f(x) \cdot f'(x) \, dx = [f(x)]^2/2 + C$
- $\int f'(x)/f(x) \, dx = \ln|f(x)| + C$

**Example (substitution):**  $\int (2x+3)^5 \, dx$ . Let  $u = 2x+3$ ,  $du = 2dx \Rightarrow dx = du/2$ .  
 $= 1/2 \int u^5 \, du = u^6/12 + C = (2x+3)^6/12 + C$ .

**Example (f'/f):**  $\int (2x)/(x^2+1) \, dx = \ln|x^2+1| + C$ .

## Integration by Parts

For product of two functions:

$$\int u \, dv = uv - \int v \, du$$

**LIATE rule** (choose u as the function that comes first): Logarithmic, Inverse trigonometric, Algebraic, Trigonometric, Exponential.

**Example:**  $\int x \sin x \, dx$ . Let  $u = x$  (algebraic),  $dv = \sin x \, dx \Rightarrow du = dx$ ,  $v = -\cos x$ .  
 $= x(-\cos x) - \int (-\cos x) \, dx = -x \cos x + \sin x + C$ .

**⚠ Common mistake:** Choosing the wrong u can make the integral more complex. Use LIATE.



## Definite Integrals

A **definite integral** has limits of integration:

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$$

**Example:**  $\int_0^1 x^2 dx = [x^3/3]_0^1 = 1/3 - 0 = 1/3.$

**Example (area):** Area bounded by  $y = x^2$ ,  $x$ -axis,  $x=0$ ,  $x=2$  is  $\int_0^2 x^2 dx = [x^3/3]_0^2 = 8/3$  sq. units.



**Engineering Application:** Definite integrals compute areas, volumes, work done, and centre of mass.

## Differential Equations

**Order:** The highest derivative present.

**Degree:** The exponent of the highest derivative (after removing radicals).

**Example:**  $d^2y/dx^2 + 3(dy/dx) + 2y = 0 \rightarrow$  Order 2, Degree 1.

**Variable separable:**  $dy/dx = f(x)g(y) \Rightarrow \int dy/g(y) = \int f(x) dx.$

**Example:**  $dy/dx = xy \Rightarrow dy/y = x dx \Rightarrow \ln|y| = x^2/2 + C \Rightarrow y = C e^{(x^2/2)}.$

**Linear first-order:**  $dy/dx + P(x)y = Q(x).$

Integrating factor:  $I.F. = e^{(\int P dx)} \cdot y \cdot I.F. = \int Q \cdot I.F. dx + C$

**Example:**  $dy/dx + 2y = x.$   $P = 2$ ,  $I.F. = e^{(\int 2 dx)} = e^{(2x)}.$   $y e^{(2x)} = \int x e^{(2x)} dx + C.$  Solve to get  $y = (2x-1)/4 + C e^{(-2x)}.$

### Definite Integral Calculator (polynomial)

Compute  $\int_a^b x^n dx$ . Enter a, b, and n.

a (lower limit)

0

b (upper limit)

2

n (exponent)

2

$$\int_a^b x^n dx = 2.666667 \text{ (a=0, b=2, n=2)}$$

## MODULE IV Numerical Methods & Vector Algebra

16 hours (L:12 + T:4)

CO4

Bloom: Apply

### Learning Goals

- Understand interpolation and Lagrange's method
- Apply Simpson's 1/3rd rule
- Distinguish scalars and vectors
- Compute dot and cross products
- Find magnitude and unit vectors

### Key Facts

- Lagrange interpolation:  $L(x) = \sum y_i \cdot \ell_i(x)$
- Simpson's 1/3rd:  $\int_a^b f(x)dx \approx h/3 [f_0 + f_n + 4\sum_{\text{odd}} + 2\sum_{\text{even}}]$
- $a \cdot b = |a||b| \cos \theta$
- $a \times b = |a||b| \sin \theta \cdot \hat{n}$

## Interpolation & Lagrange's Method

**Interpolation** is the process of estimating unknown values between known data points.

**Lagrange's method** constructs a polynomial that passes through all given points:

$$L(x) = \sum_{i=0}^n y_i \cdot \ell_i(x), \text{ where } \ell_i(x) = \prod_{j=0, j \neq i}^n (x - x_j) / (x_i - x_j)$$

**Example:** Given (1,2), (2,3), (3,5). Find y at x=2.5.

$$\ell_0 = (x-2)(x-3) / ((1-2)(1-3)) = (x-2)(x-3)/2$$

$$\ell_1 = (x-1)(x-3) / ((2-1)(2-3)) = -(x-1)(x-3)$$

$$\ell_2 = (x-1)(x-2) / ((3-1)(3-2)) = (x-1)(x-2)/2$$

$$L(2.5) = 2 \cdot \ell_0 + 3 \cdot \ell_1 + 5 \cdot \ell_2. \text{ Compute: } \ell_0 = (0.5)(-0.5)/2 = -0.125, \\ \ell_1 = -(1.5)(-0.5) = 0.75, \ell_2 = (1.5)(0.5)/2 = 0.375. L = \\ 2(-0.125) + 3(0.75) + 5(0.375) = -0.25 + 2.25 + 1.875 = 3.875.$$



**Tip:** Lagrange's method avoids solving systems of equations but can be computationally intensive for large n.

## Numerical Integration — Simpson's 1/3rd Rule

For approximating  $\int_a^b f(x) dx$  with n even subintervals of width  $h = (b-a)/n$ :

$$\int_a^b f(x) dx \approx (h/3) [f_0 + f_n + 4(f_1 + f_3 + \dots + f_{n-1}) + 2(f_2 + f_4 + \dots + f_{n-2})]$$

**Example:** Approximate  $\int_0^2 x^2 dx$  using n=4.  $h = 0.5$ .  $f_0=0$ ,  $f_1=0.25$ ,  $f_2=1$ ,  $f_3=2.25$ ,  $f_4=4$ .  $\approx (0.5/3)[0+4 + 4(0.25+2.25) + 2(1)] = (1/6)[4 + 4(2.5) + 2] = (1/6)[4+10+2] = 16/6 = 2.6667$ . Exact:  $8/3 = 2.6667$ .



**Common mistake:** n must be even for Simpson's 1/3rd rule. If n is odd, use Simpson's 3/8th rule or composite methods.

## Scalars and Vectors

- **Scalar:** A quantity with only magnitude (e.g., mass, temperature).
- **Vector:** A quantity with both magnitude and direction (e.g., velocity, force).

**Unit vectors:**  $\hat{i}, \hat{j}, \hat{k}$  along x, y, z axes.

**Position vector:**  $\mathbf{r} = x \hat{i} + y \hat{j} + z \hat{k}$ .

**Malayalam support:** Scalar-മാത്ര മാനിത (ഉദാ: mass). Vector-  
മാത്ര മാനിത-ഉപദിഷ്ട (ഉദാ: velocity). Unit vectors  $\hat{i}, \hat{j}, \hat{k}$  axes-  
പരസ്പരം ലംബമാണ്.

## Magnitude and Unit Vector

**Magnitude:**  $|\mathbf{v}| = \sqrt{x^2 + y^2 + z^2}$  for  $\mathbf{v} = x \hat{i} + y \hat{j} + z \hat{k}$ .

**Unit vector:**  $\hat{v} = \mathbf{v} / |\mathbf{v}|$  (direction of  $\mathbf{v}$ ).

**Example:**  $\mathbf{v} = 3\hat{i} + 4\hat{j}$ .  $|\mathbf{v}| = 5$ .  $\hat{v} = (3/5)\hat{i} + (4/5)\hat{j}$ .

## Vector Addition, Subtraction, Scalar Multiplication

- **Addition:**  $\mathbf{v} + \mathbf{w} = (v_1 + w_1)\hat{i} + (v_2 + w_2)\hat{j} + (v_3 + w_3)\hat{k}$
- **Subtraction:**  $\mathbf{v} - \mathbf{w} = (v_1 - w_1)\hat{i} + (v_2 - w_2)\hat{j} + (v_3 - w_3)\hat{k}$
- **Scalar multiplication:**  $c \cdot \mathbf{v} = (c v_1)\hat{i} + (c v_2)\hat{j} + (c v_3)\hat{k}$

**Example:**  $\mathbf{v} = 2\hat{i} + 3\hat{j}$ ,  $\mathbf{w} = 1\hat{i} - 2\hat{j}$ .  $\mathbf{v} + \mathbf{w} = 3\hat{i} + 1\hat{j}$ ,  $\mathbf{v} - \mathbf{w} = 1\hat{i} + 5\hat{j}$ ,  $3\mathbf{v} = 6\hat{i} + 9\hat{j}$ .

## Dot Product

$$\mathbf{v} \cdot \mathbf{w} = |\mathbf{v}| |\mathbf{w}| \cos \theta \quad \cdot \quad \mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + v_3 w_3$$

- **Angle:**  $\cos \theta = (\mathbf{v} \cdot \mathbf{w}) / (|\mathbf{v}| |\mathbf{w}|)$
- **Perpendicular:**  $\mathbf{v} \perp \mathbf{w} \Leftrightarrow \mathbf{v} \cdot \mathbf{w} = 0$

**Example:**  $\mathbf{v} = 2\hat{i} + 3\hat{j}$ ,  $\mathbf{w} = 1\hat{i} - 2\hat{j}$ .  $\mathbf{v} \cdot \mathbf{w} = 2(1) + 3(-2) = 2 - 6 = -4$ .  
 $|\mathbf{v}| = \sqrt{13}$ ,  $|\mathbf{w}| = \sqrt{5}$ ,  $\cos \theta = -4 / (\sqrt{13} \cdot \sqrt{5}) \approx -0.496$ .

## Cross Product

$$\mathbf{v} \times \mathbf{w} = |\mathbf{v}| |\mathbf{w}| \sin \theta \cdot \hat{n} \quad \cdot \quad \mathbf{v} \times \mathbf{w} = (v_2 w_3 - v_3 w_2) \hat{i} - (v_1 w_3 - v_3 w_1) \hat{j} + (v_1 w_2 - v_2 w_1) \hat{k}$$

- **Parallel:**  $\mathbf{v} \parallel \mathbf{w} \Leftrightarrow \mathbf{v} \times \mathbf{w} = \mathbf{0}$

**Example:**  $\mathbf{v} = 2\hat{i} + 3\hat{j}$ ,  $\mathbf{w} = 1\hat{i} - 2\hat{j}$ .  $\mathbf{v} \times \mathbf{w} = (3 \cdot 0 - 0 \cdot (-2)) \hat{i} - (2 \cdot 0 - 0 \cdot 1) \hat{j} + (2 \cdot (-2) - 3 \cdot 1) \hat{k} = -7\hat{k}$ .

 **Engineering Application:** Cross product gives torque ( $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ ) and is used in magnetism and rotational dynamics.



## Vector Calculator

Enter components of two vectors (a, b) and compute dot product, cross product, magnitudes, and angle.

$a_1$

2

$a_2$

3

$a_3$

0

$b_1$

1

$b_2$

-2

$b_3$

0

$a \cdot b = -4.000$  |  $a \times b = (0.000, 0.000, -7.000)$  |  $|a| = 3.606$  |  $|b| = 2.236$  |  $\theta = 119.74^\circ$



## Formula Bank

$$z = a + ib, i^2 = -1$$

Complex number

$$|z| = \sqrt{a^2 + b^2}$$

Modulus

$$\theta = \tan^{-1}(b/a)$$

Argument

$$z = r(\cos \theta + i \sin \theta)$$

Polar form

$$P(E) = n(E)/n(S)$$

Probability

$$\bar{x} = \Sigma x / n$$

Mean

$$\int x^n dx = x^{n+1}/(n+1) + C$$

Power rule

$$\int u dv = uv - \int v du$$

Integration by parts

$$\int_a^b f(x) dx = F(b) - F(a)$$

Definite integral

$$I.F. = e^{\int P dx}$$

Linear D.E.

$$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + v_3 w_3$$

Dot product

$$\mathbf{v} \times \mathbf{w} = (v_2 w_3 - v_3 w_2) \hat{i} - \dots$$

Cross product



## Diagram Library for Rapid Revision



### Argand Plane

Module 1 — Argand Plane showing complex number representation.



### Probability Tree

Visualise outcomes for coin/die experiments via tree diagrams (conceptual).



### Vector Geometry

Visualise dot and cross products in 3D space. Module 4



## Self-Learning Activities (Official)

#### Week 1-4 · 16 h

##### Module I — Complex Numbers

- Find conjugate, modulus, amplitude, polar form
- Problems on equality, addition, subtraction
- Multiply and divide complex numbers
- Applications in engineering areas

#### Week 5-8 · 15 h

##### Module II — Probability & Statistics

- Problems with coins, die, cards
- Find mean, median, mode
- Frequency table problems
- Applications in engineering

#### Week 9-12 · 14 h

##### Module III — Integral Calculus

- Find integrals of functions
- Evaluate definite integrals
- Solve first-order differential equations
- Applications in engineering

#### Week 13-15 · 15 h

##### Module IV — Numerical Methods & Vectors

- Interpolation and Simpson's 1/3rd rule problems
- Vector addition, subtraction, scalar multiplication
- Dot product and cross product
- Area and work done problems

**Total self-learning hours:** 60 (as per syllabus)



## Assessment Methodology



## CIE / ESE Breakup

| Mode         | Attendance          | CA1 | CA2 | CA3 | CA4 | Series Exam (2 modules) | Series Exam (2 modules) | ESE  |
|--------------|---------------------|-----|-----|-----|-----|-------------------------|-------------------------|------|
| Duration     | —                   | 2h  | 2h  | 2h  | 2h  | 2h                      | 2h                      | 2.5h |
| Exam mark    | —                   | 15  | 15  | 15  | 15  | 15                      | 15                      | 60   |
| Converted to | 5                   | 15  |     |     |     | 20                      |                         | 60   |
| <b>Total</b> | 40 (CIE) + 60 (ESE) |     |     |     |     |                         |                         |      |

### Series Examination Pattern

**Duration:** 2 hours

**Total marks:** 50 (converted to 20)

- Part A:  $2 \times 1$  mark
- Part B:  $2 \times 3$  marks
- Part C:  $2 \times 7$  marks (with choice)

### End Semester Examination

**Duration:** 2.5 hours

**Total marks:** 60

- Part A: 1 mark  $\times$  12 questions (2 per module)
- Part B: 3 marks  $\times$  6 questions (2 out of 3 per module)
- Part C: 7 marks  $\times$  6 questions (1 set with choice per module)

## ? Module-wise Questions with Answers

### Module I — Complex Numbers

**Q1: Find the modulus and argument of  $z = 1 + i$ .**▼

$$|z| = \sqrt{(1^2+1^2)} = \sqrt{2}. \theta = \tan^{-1}(1/1) = 45^\circ.$$

**Q2: Express  $z = 2 - 2i$  in polar form.**▼

$$r = \sqrt{(4+4)} = 2\sqrt{2}. \theta = -45^\circ. z = 2\sqrt{2}(\cos(-45^\circ) + i \sin(-45^\circ)).$$

**Q3: If  $z_1 = 1+2i$  and  $z_2 = 3-i$ , find  $z_1 \times z_2$ .**▼

$$(1+2i)(3-i) = 3-i+6i-2i^2 = 3+5i+2 = 5+5i.$$

**Q4: What is the conjugate of  $3 - 4i$ ?**▼

Conjugate is  $3 + 4i$ .

## Module II — Probability & Statistics

**Q1: A coin is tossed twice. Find the probability of getting exactly one head.**▼

$$S = \{HH, HT, TH, TT\}. n(S)=4. E = \{HT, TH\}. n(E)=2. P = 2/4 = 1/2.$$

**Q2: Find the mean of 3, 5, 7, 7, 9.**▼

$$\text{Mean} = (3+5+7+7+9)/5 = 31/5 = 6.2.$$

**Q3: What is the median of 2, 4, 6, 8?**▼

$$n=4 \text{ (even)}. \text{Median} = (4+6)/2 = 5.$$

**Q4: A die is rolled. Find P(prime number).**▼

$$S = \{1,2,3,4,5,6\}. \text{Primes} = \{2,3,5\}. P = 3/6 = 1/2.$$

## Module III — Integral Calculus

**Q1: Evaluate  $\int 3x^2 \, dx$ .**▼

$$\int 3x^2 \, dx = 3 \cdot (x^3/3) + C = x^3 + C.$$

**Q2: Evaluate  $\int_0^1 x \, dx$ .**▼

$$[x^2/2]_0^1 = 1/2 - 0 = 1/2.$$

**Q3: Solve  $dy/dx = 2x$ .**▼

$$dy = 2x \, dx \Rightarrow \int dy = \int 2x \, dx \Rightarrow y = x^2 + C.$$

**Q4: Find the order and degree of  $d^2y/dx^2 + 3(dy/dx) + 2y = 0$ .**▼

Order = 2 (highest derivative), Degree = 1 (exponent of highest derivative).

## Module IV — Numerical Methods & Vector Algebra

**Q1: Given (1,2), (2,3), (3,5), find y at x=2.5 using Lagrange's method.**▼

$$L(2.5) = 2 \cdot \ell_0 + 3 \cdot \ell_1 + 5 \cdot \ell_2 = 3.875 \text{ (as worked in section).}$$

**Q2: Approximate  $\int_0^1 x^2 \, dx$  using Simpson's 1/3rd with n=2.**▼

$$h = 0.5. \, f_0=0, \, f_1=0.25, \, f_2=1. \, \approx (0.5/3)[0+1+4(0.25)] = (1/6)[1+1] = 1/3 = 0.3333.$$

**Q3: Find the magnitude of  $\mathbf{v} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$ .**▼

$$|\mathbf{v}| = \sqrt{9+16} = 5.$$

**Q4: Compute the dot product of  $\mathbf{a} = (2,3)$  and  $\mathbf{b} = (1,-2)$ .**▼

$$\mathbf{a} \cdot \mathbf{b} = 2(1)+3(-2) = 2-6 = -4.$$



## Practice Model Paper

**Note:** This is a practice paper based on the official pattern, not an official SITTTR model question paper.

**Course 2001A — Engineering Mathematics – I**

**Duration:** 2.5 hours · **Total marks:** 60

*All modules are covered. Answer all parts.*

### Part A (1 mark each — 12 questions, 2 per module)

- **Q1.** What is the conjugate of  $2 + 5i$ ? (M1)

- **Q2.** Find  $|3 - 4i|$ . (M1)
- **Q3.** A coin is tossed once. What is  $P(H)$ ? (M2)
- **Q4.** Define sample space. (M2)
- **Q5.**  $\int 2x \, dx = ?$  (M3)
- **Q6.** What is the order of  $d^2y/dx^2 + y = 0$ ? (M3)
- **Q7.** What is a unit vector? (M4)
- **Q8.** When are two vectors perpendicular? (M4)
- **Q9-12.** (Additional questions covering all modules)

**Part B (3 marks each — 6 questions, 2 out of 3 per module)**

- **M1:** Express  $1 - i$  in polar form. *OR* Find modulus and argument of  $-1 + i$ .
- **M2:** Two coins are tossed. Find  $P(\text{at least one head})$ . *OR* Find mean/median/mode of 2,3,3,5,6.
- **M3:** Evaluate  $\int x \sin x \, dx$ . *OR* Solve  $dy/dx = 2xy$ .
- **M4:** Given (1,2), (2,4), (3,6), find  $y$  at  $x=2.5$  using Lagrange. *OR* Find  $a \cdot b$  and  $a \times b$  for  $a=(1,2,0)$ ,  $b=(0,1,1)$ .

**Part C (7 marks each — 6 questions, 1 set with choice per module)**

- **M1:** If  $z_1 = 2+3i$  and  $z_2 = 1-2i$ , find  $z_1+z_2$ ,  $z_1-z_2$ ,  $z_1 \cdot z_2$ , and  $z_1/z_2$ . *OR* Express  $z = 2+2i$  in polar form and find  $z^3$ .
- **M2:** A card is drawn from a deck. Find  $P(\text{heart})$ ,  $P(\text{ace})$ ,  $P(\text{heart or ace})$ . *OR* Given frequency distribution, compute mean, median, mode.
- **M3:** Evaluate  $\int_0^2 (x^2+2x) \, dx$ . Solve  $dy/dx + 2y = e^x$ . *OR* Find area under  $y = x^2$  from  $x=0$  to  $x=2$ .
- **M4:** Use Simpson's 1/3rd to approximate  $\int_0^2 x^3 \, dx$  with  $n=4$ . Find  $a \times b$  for  $a=(2,1,0)$ ,  $b=(0,3,4)$ . *OR* Lagrange interpolation problem with 4 points.

## Quick Revision

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## Module I — Complex Numbers

- $z = a + ib, i^2 = -1$
- $|z| = \sqrt{a^2+b^2}, \theta = \tan^{-1}(b/a)$
- Polar:  $z = r(\cos \theta + i \sin \theta)$
- Operations: add real/imag separately

## Module II — Probability & Stats

- $P(E) = n(E)/n(S)$
- Mean =  $\Sigma x/n$ , Median = middle, Mode = most frequent
- Sample space for coins/dice/cards

## Module III — Integral Calculus

- $\int x^n dx = x^{n+1}/(n+1) + C$
- $\int u dv = uv - \int v du$
- Definite:  $\int_a^b = F(b) - F(a)$
- D.E.: variable separable & linear

## Module IV — Num Methods & Vectors

- Lagrange:  $L(x) = \Sigma y_i \cdot \ell_i(x)$
- Simpson's:  $h/3 [f_0 + f_n + 4\Sigma \text{odd} + 2\Sigma \text{even}]$
- $a \cdot b = |a||b| \cos \theta$
- $a \times b = |a||b| \sin \theta \cdot \hat{n}$

## ⚠ Common Mistakes — Exam Checklist

- ✓ In complex division, multiply by conjugate of denominator
- ✓ In probability, check if order matters (permutations vs combinations)
- ✓ In integration, always add +C for indefinite integrals
- ✓ In Simpson's rule, n must be even
- ✓ In vectors, cross product gives a vector, dot product gives a scalar
- ✓ In differential equations, identify order and degree correctly

# Glossary

| Term                  | Meaning                                                         |
|-----------------------|-----------------------------------------------------------------|
| Complex number        | $z = a + ib$ , where $i^2 = -1$                                 |
| Modulus               | $ z  = \sqrt{a^2+b^2}$ , distance from origin                   |
| Argument              | $\theta = \tan^{-1}(b/a)$ , angle with real axis                |
| Polar form            | $z = r(\cos \theta + i \sin \theta)$                            |
| Sample space          | Set of all possible outcomes                                    |
| Probability           | $P(E) = n(E)/n(S)$                                              |
| Mean                  | Average = $\Sigma x/n$                                          |
| Median                | Middle value of ordered data                                    |
| Mode                  | Most frequent value                                             |
| Integration           | Reverse process of differentiation                              |
| Definite integral     | $\int_a^b f(x) dx = F(b) - F(a)$                                |
| Differential equation | Equation involving derivatives                                  |
| Interpolation         | Estimating unknown values from known data                       |
| Lagrange's method     | Polynomial interpolation through given points                   |
| Simpson's 1/3rd rule  | Numerical integration using parabolic arcs                      |
| Scalar                | Quantity with only magnitude                                    |
| Vector                | Quantity with magnitude and direction                           |
| Dot product           | Scalar product: $a \cdot b =  a  b  \cos \theta$                |
| Cross product         | Vector product: $a \times b =  a  b  \sin \theta \cdot \hat{n}$ |

## Learning Resources

| Textbooks |                                        |                      |
|-----------|----------------------------------------|----------------------|
| Sl. No.   | Title                                  | Author / Publication |
| 1         | Mathematics for Industrial Engineering | SITTTR Kalamasser    |

| References |                                                    |                                                        |
|------------|----------------------------------------------------|--------------------------------------------------------|
| Sl. No.    | Title                                              | Author / Publication                                   |
| 1          | Higher Engineering Mathematics                     | B. S. Grewal, Khanna Publishers                        |
| 2          | Higher Engineering Mathematics                     | B. V. Ramana, Tata McGraw-Hill                         |
| 3          | A text book of Engineering Mathematics             | N. P. Bali & Manish Goyal, Laxmi Publications          |
| 4          | Schau's outlines of Discrete Mathematics (3rd Ed.) | Seymour Lipschutz & Marc Lipson, McGraw Hill           |
| 5          | Schau's outlines of Probability (3rd Ed.)          | Seymour Lipschutz & Marc Lipson, McGraw Hill           |
| 6          | Schau's outlines of Statistics (6th Ed.)           | Dr. Murray R. Spiegel & Larry J. Stephens, McGraw Hill |

**Web resources:** None officially listed in the syllabus.